COURSE CODE: CE372

COURSE NAME: Quantity Surveying & Cost Estimation (QS&CE)

CREDIT HOURS: Theory = 3
Practical = 0
Total = 3

CONTACT HOURS: Theory = 48
Practical = 0
Total = 48

PRE-REQUISITE: CE-181 Civil Engineering Drawings

MODE OF TEACHING:

Instructions: Three hours of lecture per week 100%

Practical/Laboratory Demonstration: Three hours of lecture per week 0

COURSE DESCRIPTION:

This course is designed to prepare students to handle the practical tasks of quantity surveying and cost estimation to which they will be exposed to in their real professional life. Different types of estimates are studied. The detailed estimates for each major civil works are prepared based upon real construction project documents. This course teaches the methodology, procedures and organizational techniques involved in preparing cost estimates. The course also covers rate analysis. Students are introduced with general practice in government departments for bills of quantities, measurement books, schedule of rate and specifications. Preparation of deviation orders, interim and final payments certificate are also discussed.

COURSE OBJECTIVES:

- To enable students to take off quantities from civil engineering drawings, prepare bill of quantities, store statements, deviation orders and payment certificates.

- To enable students to select appropriate estimating method from the feasibility phase up to the implementation phase of the project.
- To bring awareness in students about the ethical considerations of estimating and billings and their effect on the people and project.

DETAILS OF ASSIGNMENTS/PRACTICE

- Five assignments given to the students
- A project of about 15% weightage to be carried out by the students

RELEVANT PROGRAM LEARNING OUTCOMES (PLOs):

The course is designed so that students will achieve the PLOs:

1	Engineering Knowledge:	☑	7	Environment and Sustainability:	☐
2	Problem Analysis:	☐	8	Ethics:	☐
3	Design/Development of Solutions:	☐	9	Individual and Team Work:	☐
4	Investigation:	☐	10	Communication:	☐
5	Modern Tool Usage:	☐	11	Project Management:	☐
6	The Engineer and Society:	☐	12	Lifelong Learning:	☐

COURSE LEARNING OUTCOMES (CLOs):

Upon successful completion of the course, the student will demonstrate competency by being able to:

S. No	Outcomes	Domain	Level of Learning	PLO
1.	Understand and comprehend the quantity take-offs using standard engineering methods estimates, their types and procedural requirements	Cognitive	3	1
2.	*Perform* basic building construction measurements and *prepare* cost and material estimates by studying and scrutinizing quantities of various construction activities	Cognitive	4	1

PRACTICAL APPLICATION:

The estimation of cost and resources are prime consideration for all projects. Quantity surveying and cost estimation is applicable right from the feasibility phase up to the completion phase of any project. Accurate forecasting of work not only gives confidence to the client to initiate the project but also gives fair grounds to the contractors to bid competitively. Comprehensive estimates and exact measurements of work at site forms the basis for genuine payments and results in minimizing sick projects.

TOPICS COVERED:

Week/ Lecture	Topic
1	Introduction to cost estimating and design process. Types of cost estimates.
2-3	Design & Estimating Process
4-5	Types of Cost Estimates
6-7	Preconstruction Services and Estimate
8-9	BOQs & MBs
10-11	Quantity Take Off
12-13	Pricing Civil Works
14	Estimating General Conditions
15-16	Estimating Subcontractor Works

REFERENCE BOOKS: (including Textbooks and Reference books)

1. MES Schedule of Rates 2009 (Ref Book)
2. Professional Practice (Estimation & Valuation) by Roshan Namavati, Union Book Stall, M,A Jinnah Road Karachi
3. Estimating Construction Costs by Robert L. Peurifoy and Garold D. Oberlender (2008) McGraw-Hill Book Company, New York
4. Construction Cost estimating Process and Practices by Holm, Len; Schaufelberger, John; Griffin, Dennis & Cole Thomas (2005) Pearson Prentice Hall, Upper Saddle River, New Jersey
5. Estimating in Building Construction (7thed). Frank R. Dagonstino, Steven J. Peterson (2011). Pearson Prentice Hall. Upper Saddle River, New Jersey
6. Estimating and Costing by B.N. Dutta

ASSESMENT SYSTEM:

1. **CLOs Assessment**

Cognitive	Psychomotor	Affective
Spreadsheet	Rubrics	Rubrics

2. **Relative Grading**

Theoretical / Instruction			100%
	Assignments 10%		
	Quizzes 10%		
	OHT Exams 30%		
	End Semester Exam 50%		
Total			100%

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